

Monitoring & Incident Response Plan

Enhancing Security in Messaging Applications

Spinnaker Analytics

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1. Introduction

1.1 Purpose

- This document defines the monitoring, detection, alerting, and incident response procedures implemented to protect the messaging application against cyber threats and operational security incidents.
- The objective is to ensure timely identification, analysis, containment, eradication, and recovery from security events while maintaining the confidentiality, integrity, and availability of messaging services.

1.2 Objectives

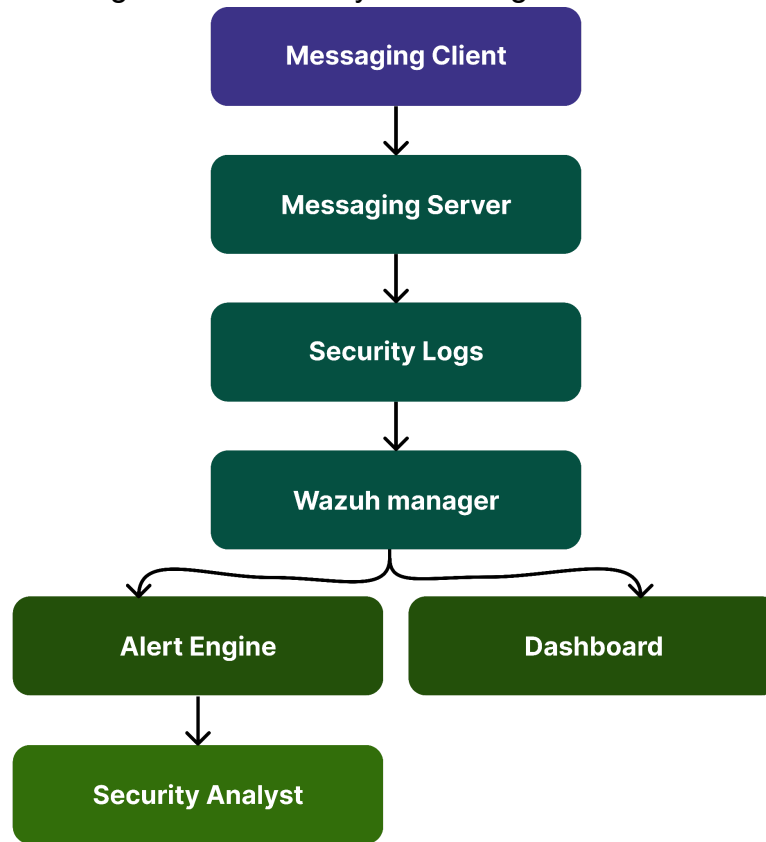
- Continuous security monitoring
- Early threat detection
- Centralized log collection
- Incident response procedures
- Alert management
- Security reporting
- Continuous improvement

2. Security Monitoring Architecture

The monitoring solution collects logs from client applications, messaging servers, operating systems, network devices, and security tools.

2.1 Security Monitoring Architecture

Figure 31 – Security Monitoring Architecture



2.2 Monitoring Components

Component	Purpose
Wazuh Manager	Central log collection
Wazuh Agent	Endpoint monitoring
Suricata IDS/IPS	Network threat detection
OpenSearch Dashboard	Log visualization
OpenVAS	Vulnerability assessment

3. Security Event Logging

3.1 Logged Events

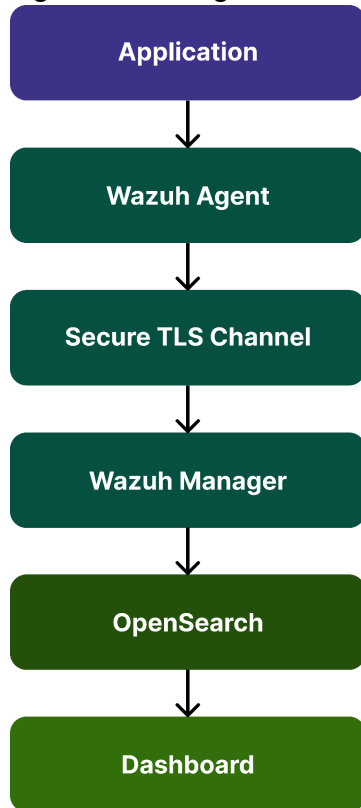
Event	Logged
User Login	Yes
Failed Login	Yes
Password Reset	Yes
MFA Failure	Yes
File Upload	Yes
Voice Call Initiation	Yes
Administrative Actions	Yes
Security Configuration Changes	Yes

3.2 Log Retention

Log Type	Retention
Authentication Logs	90 Days
Application Logs	180 Days
Security Alerts	1 Year
Incident Reports	1 Year

3.3 Log Collection

Figure 32 – Log Collection Workflow



4. Threat Detection & Monitoring

4.1 Detection Sources

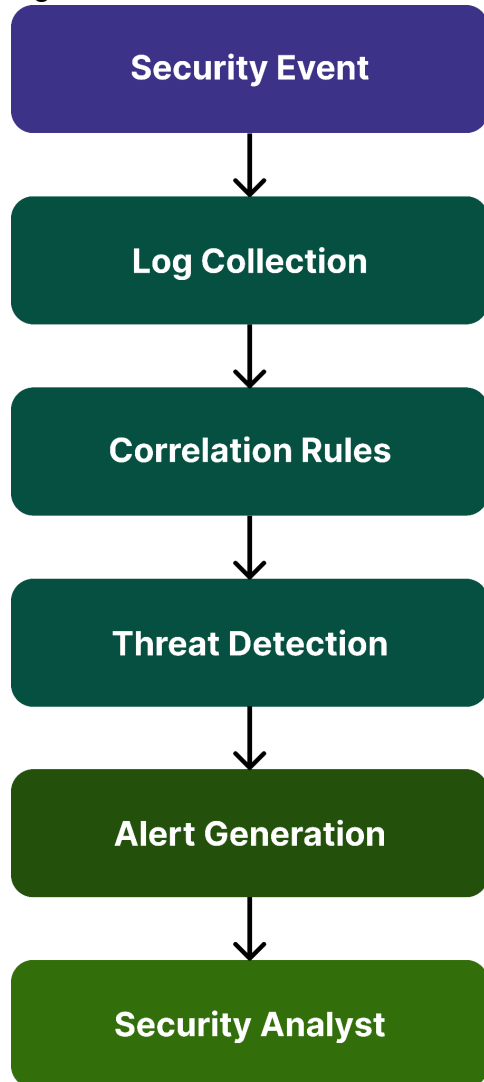
Source	Threat Detected
Wazuh	Host-based events
Suricata	Network attacks
OpenVAS	Vulnerability exposure
System Logs	Authentication anomalies
Application Logs	Unauthorized activity

4.2 Alert Severity Levels

Severity	Response
Critical	Immediate response
High	Within 1 Hour
Medium	Within 4 Hours
Low	Next business day

4.3 Threat Detection

Figure 33 – Threat Detection Workflow

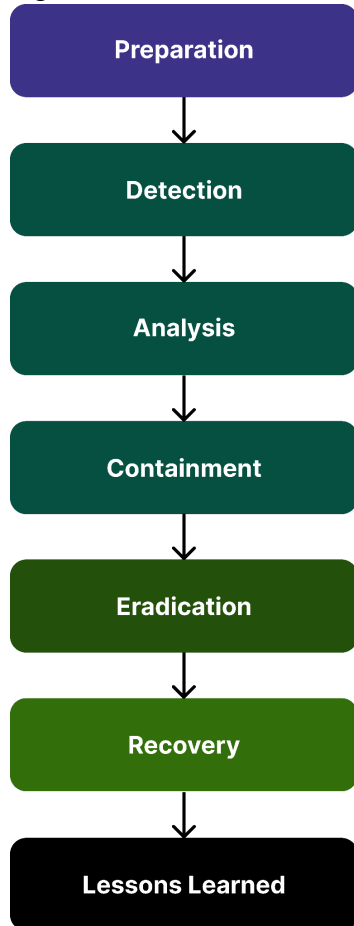


5. Incident Response Framework

The incident response process follows the NIST SP 800-61 Rev. 2 lifecycle.

5.1 Incident Response

Figure 34 – Incident Response Lifecycle



5.1 Incident Categories

Category	Example
Authentication	Brute-force login
Malware	Infected endpoint
Data Leakage	Unauthorized disclosure
Network Attack	DoS / Scanning
Insider Threat	Unauthorized administrative activity

5.3 Escalation Matrix

Severity	Escalated To	Response Time
Critical	SOC Manager	Immediate
High	Security Analyst	1 Hour
Medium	System Administrator	4 Hours
Low	IT Operations	Next Business Day

6. Incident Handling Procedures

6.1 Incident Workflow

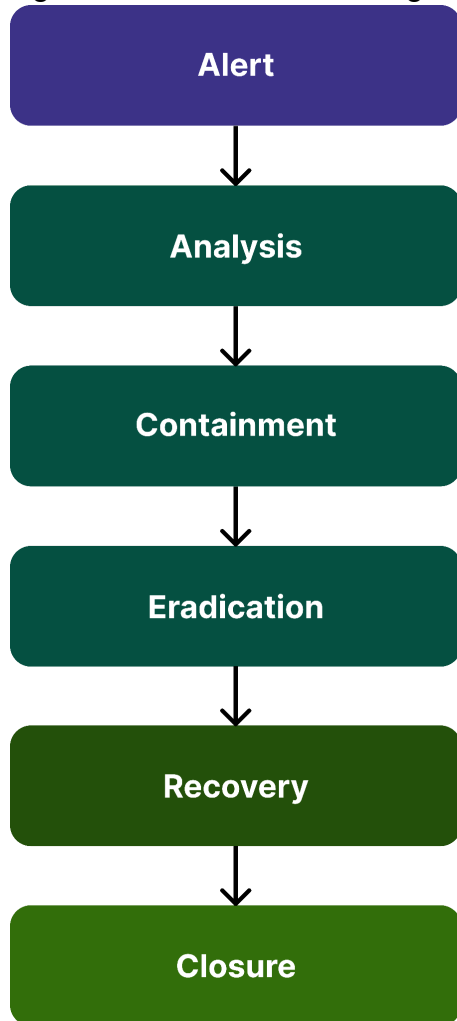
- **Step 1 – Detection**
 - Alert generated
 - Incident ticket created
- **Step 2 – Analysis**
 - Validate alert
 - Determine severity
- **Assess impact**
 - Step 3 – Containment
 - Isolate affected systems
 - Block malicious IPs
 - Disable compromised accounts
- **Step 4 – Eradication**
 - Remove malicious artifacts
 - Patch vulnerabilities
 - Reset credentials
- **Step 5 – Recovery**
 - Restore services
 - Verify system integrity
 - Resume operations
- **Step 6 – Post-Incident Review**
 - Root cause analysis
 - Lessons learned
 - Update security controls

6.2 Incident Classification

Incident Type	Priority
Data Breach	Critical
Malware Infection	High
Unauthorized Login	High
Configuration Error	Medium
Failed Login Attempts	Low

6.3 Incident Handling

Figure 35 – Incident Handling Workflow



7.Security Monitoring Validation

7.1 Validation Tools

Tool	Purpose
Wazuh	Alert generation
Suricata	IDS validation
OpenVAS	Vulnerability scanning
Wireshark	Traffic inspection
Burp Suite	Web/API validation

7.2 Test Cases

Test ID	Scenario	Expected Result
MON-01	Failed login attempts	Alert generated
MON-02	Port scan	IDS alert
MON-03	Malware signature	Detection event
MON-04	Unauthorized privilege change	High severity alert
MON-05	Vulnerability scan	Findings recorded
MON-06	Log forwarding	Dashboard updated

7.3 Expected Evidence

- Wazuh dashboard screenshots
- Suricata alerts
- OpenVAS reports
- Wireshark captures
- Incident tickets

8. Security Governance & Reporting

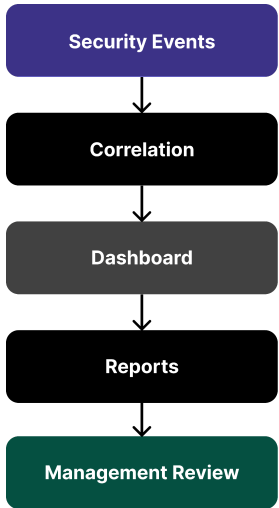
8.1 Reporting Metrics

Metric	Frequency
Security Alerts	Daily
Incident Summary	Weekly
Vulnerability Report	Monthly
Patch Compliance	Monthly
Executive Security Report	Quarterly

8.2 Key Performance Indicators (KPIs)

KPI	Target
Mean Time to Detect (MTTD)	< 15 Minutes
Mean Time to Respond (MTTR)	< 1 Hour
Critical Incident Resolution	< 24 Hours
Log Collection Availability	99.9%

8.3 Security Reporting



9. Standards Alignment

Standard	Purpose
NIST SP 800-61 Rev.2	Incident Response
NIST CSF 2.0	Security Monitoring
ISO/IEC 27035	Incident Management
MITRE ATT&CK	Threat Classification
OWASP MASVS	Mobile Security Monitoring

10. Conclusion

- This Monitoring & Incident Response Plan establishes a structured approach to detecting, analyzing, responding to, and recovering from security incidents affecting the messaging application.
- The solution combines centralized monitoring through Wazuh, network detection using Suricata, vulnerability assessment with OpenVAS, and standardized incident response procedures based on NIST guidance.
- Continuous monitoring, defined escalation paths, and measurable performance indicators improve operational resilience and reduce the impact of security events.